



Multi-Objective Optimisation of HIV Intervention Funding Allocation Across African Countries using NSGA-II

40.018 Heuristics and Systems Theory Project Report

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1 Problem Background

Following disruptions and reductions in U.S. foreign assistance, including activities supported by PEPFAR, critical gaps have emerged in HIV treatment and prevention programmes across high-burden African countries. In this project scenario, the Gates Foundation is assumed to commit a fixed budget of USD 200 million to help address these gaps. The objective is to allocate this budget across four intervention types; Antiretroviral (ARV) procurement, supply chain infrastructure, workforce training, and HIV Drug Resistance (HIVDR) surveillance so as to maximise expected Disability-Adjusted Life Years (DALYs)¹ averted over a 5-year horizon while accounting for both intervention effectiveness and country-level funding needs. Such systems have been heavily critiqued due to ethical concerns, as they prioritise impact per dollar and may therefore fail to fully reflect absolute need, equity, and rights-based access to treatment, a tension also discussed in global health debates on cost-effectiveness (Glassman, 2014).

Although the broader motivation is Africa-wide HIV funding allocation, the final implementation focuses on **46 African countries** with sufficient matched UNAIDS and IHME data. Countries with missing or insufficient HIV burden data were excluded during preprocessing. The model should therefore be interpreted as an optimisation over data-available African countries rather than a complete allocation across all 54.

With 46 countries, 4 intervention types, continuous funding decisions, soft-constraint penalties for ARV supply shortages, saturating impact functions, and bilinear interdependencies between interventions, the full optimisation problem is a large-scale non-convex nonlinear program (NLP). The presence of nonlinear diminishing returns, shortage penalties, and the bilinear ARV–Supply synergy term makes direct solver-based optimisation computationally challenging. This motivates the use of multi-objective metaheuristic approaches to approximate high-quality funding allocations across the Pareto frontier.

2 Problem Formulation

2.1 Sets

- C : set of countries, $c \in C$, $|C| = 46$. The 46 countries span the four sub-Saharan regions: 16 in West Africa, 9 in Central Africa, 16 in East Africa, and 5 in Southern Africa. Eight North African and island countries (Algeria, Egypt, Libya, Morocco, Sudan, Tunisia, Mauritius, Seychelles) were excluded due to missing UNAIDS or IHME data. A full country roster is provided in the code file.
- I : set of interventions, $I = \{\text{ARV, Supply, Workforce, HIVDR}\}$.

The final implementation therefore includes $46 \times 4 = 184$ country-intervention funding decision cells. Regional groupings are used only to organise results and are not part of the decision structure.

2.2 Decision Variables

- $x_{c,i} \in \mathbb{R}^+$: continuous funding allocated to intervention i in country c , measured in millions of USD.
- $s_c^{\text{short}} = \max(T_c - x_{c,\text{ARV}}, 0) \in \mathbb{R}^+$: the ARV funding shortfall below the minimum viable ARV threshold in country c . Defined implicitly rather than as a solver variable, and included in the objective as a soft-penalty term rather than as an equality constraint.

¹DALYs measure the overall burden of disease; one DALY represents one lost year of “healthy” life.

A candidate solution is encoded as a 46×4 allocation matrix

$$X = \begin{bmatrix} x_{1,\text{ARV}} & x_{1,\text{Supply}} & x_{1,\text{Workforce}} & x_{1,\text{HIVDR}} \\ x_{2,\text{ARV}} & x_{2,\text{Supply}} & x_{2,\text{Workforce}} & x_{2,\text{HIVDR}} \\ \vdots & \vdots & \vdots & \vdots \\ x_{46,\text{ARV}} & x_{46,\text{Supply}} & x_{46,\text{Workforce}} & x_{46,\text{HIVDR}} \end{bmatrix},$$

giving 184 continuous funding decision variables. This encoding is consistent with the implementation, in which each chromosome is a country-intervention funding matrix.

2.3 Parameters

- N_c : population living with HIV in country c , obtained from UNAIDS 2024 PLHIV estimates. Used to scale intervention impact by country-level need.
- T_c : minimum viable ARV funding threshold in country c , in USD millions:

$$T_c = \frac{33.24 \cdot N_c}{10^6}.$$

By 2023 ART Clinical Guidelines, standard first-line ART for adults in sub-Saharan Africa is a combination of Tenofovir, Lamivudine, and Dolutegravir (TLD). At 2026 GHSC-PSM contract pricing, a 90-tablet TLD bottle costs USD 8.31, giving an annual per-patient drug cost of USD 33.24. To avoid double-counting, logistics and supply-chain markup are not reflected in T_c , but is modelled separately through the Supply intervention below.

- $B_{c,i}$: maximum DALY-averting potential of intervention i in country c . The parameter uses IHME Global Burden of Disease HIV/AIDS DALY estimates and is split across interventions using operational-dependency weights: ARV 65%, Supply 15%, Workforce 15%, HIVDR 5%. These weights are established as such due to the following reasons: (i) ARVs are the direct therapeutic input. Without ARV procurement, the other three interventions would have limited immediate treatment value. (ii) Supply chains and Workforce are the enabling infrastructure without which drugs cannot reach patients or be prescribed safely (iii) HIVDR surveillance protects long-run programme effectiveness through resistance detection.
- $U_{c,i}$: maximum absorbable funding for intervention i in country c , in USD millions:

$$U_{c,i} = \frac{N_c \cdot k_i}{10^6}, \quad k = (33.24, 16.62, 8.31, 20.00) \text{ USD/patient/year}.$$

The per-patient cost anchors k_i are reference cost assumptions used to scale each intervention by country-level PLHIV population. They do not represent exact country-specific costs, but provide a consistent basis for absorption-capacity calculation. Their derivations are:

- **ARV (USD 33.24 / patient / year)**. Derived from GHSC-PSM 2026 contract pricing: a 90-tablet TLD bottle costs USD 8.31, and a one-pill-per-day regimen requires approximately four bottles per patient per year, giving $4 \times 8.31 = 33.24$. This is the raw annual drug cost per patient at contract prices.
- **Supply Chain (USD 16.62 / patient / year)**. Modelled as a 50% logistics markup on the raw ARV cost ($0.5 \times 33.24 = 16.62$). This captures warehousing, transport, and last-mile distribution costs. It is modelled separately from T_c to avoid double-counting supply-chain overhead in the ARV threshold.
- **Workforce (USD 8.31 / patient / year)**. Modelled as 25% of raw ARV cost ($0.25 \times 33.24 = 8.31$), reflecting personnel cost for prescribing, monitoring, adherence support, and dispensing. This is a heuristic anchor consistent with sub-Saharan health-costing benchmarks in which workforce support is roughly a quarter of drug procurement cost.

- **HIVDR (USD 20.00 / patient / year)**. An approximate per-sample cost for HIV drug-resistance testing and surveillance, based on published assay-cost ranges for viral genotyping in low-resource settings.
- $\beta_{c,i} = B_{c,i}/U_{c,i}$: capacity-normalised impact coefficient, representing the average DALY potential per USD million up to the absorbable funding ceiling $U_{c,i}$. Used for greedy ranking and the reduced linearised Gurobi benchmark; the full NSGA-II objective evaluates impact using the saturating response function below (Eq. 1).
- κ_i : intervention-specific saturation rate governing diminishing returns in Eq. 1. A larger κ_i means the intervention reaches its impact ceiling faster as dollars per PLHIV increase; a smaller κ_i means impact grows more gradually. We set $\kappa = (0.02, 0.06, 0.12, 0.05)$ for (ARV, Supply, Workforce, HIVDR). These values are heuristically anchored to the inverse of the per-patient cost anchors above:

$$\kappa_{\text{Supply}} \approx 1/16.62 = 0.060, \quad \kappa_{\text{Workforce}} \approx 1/8.31 = 0.120, \quad \kappa_{\text{HIVDR}} = 1/20.00 = 0.050.$$

The rationale is that cheaper per-patient interventions saturate faster as a fixed dollar amount covers a larger fraction of the at-risk population. ARV is deliberately set to 0.02 rather than $1/33.24 \approx 0.030$ to reflect that ARV impact saturates more slowly than the raw drug cost alone suggests. Sustained ARV effectiveness depends on continuous treatment coverage, patient adherence, dispensing infrastructure, and system delivery, none of which is captured by a single-year drug purchase. The lower κ_{ARV} conservatively models this slower saturation.

- γ_c : additive ARV–Supply synergy coefficient. This is calibrated as

$$\gamma_c = 0.001 + 0.009 \cdot \frac{B_{c,\text{total}}}{\max_c B_{c,\text{total}}},$$

a burden-normalised parameter in the range $[0.001, 0.010]$. This substitutes for the ART-coverage-gap parameterisation intended in the original proposal, because per-country ART coverage data was not available in the assembled dataset. In the implemented model, $x_{c,\text{ARV}}$ and $x_{c,\text{Supply}}$ are measured in USD millions, so the additive term $\gamma_c x_{c,\text{ARV}} x_{c,\text{Supply}}$ is small relative to the direct saturating DALY response and ARV shortage penalty. For the largest-allocation countries, its maximum value is on the order of tens of DALYs, compared with direct ARV impact on the order of 10^6 DALYs. It should therefore be interpreted as a minor interaction proxy or tie-breaking bonus favouring joint ARV–Supply funding, rather than a material driver of allocation decisions.

- $P = 1000$: DALYs penalised per USD million of ARV shortfall s_c^{short} . This is calibrated so that the marginal DALY loss from failing to meet T_c by USD 1M is larger than the marginal DALY gain from spending that USD 1M on any non-ARV intervention in a typical country. This ensures ARV threshold satisfaction is preferred wherever absorption capacity permits, while remaining a soft constraint that allows for infeasibility under the USD 200M / USD 872M underfunding regime.
- $\alpha = 0.5$: workforce-to-supply absorption ratio, matching the ratio of per-patient workforce cost (USD 8.31) to per-patient supply cost (USD 16.62). This structural coefficient enforces that workforce spend cannot exceed half of the supporting supply-chain spend (plus baseline).

2.4 Saturating Impact Response Function

The expected DALYs averted by each country–intervention pair follow a concave saturating response function:

$$\text{DALY}_{c,i}(x_{c,i}) = B_{c,i} \left(1 - \exp \left(-\kappa_i \cdot \frac{x_{c,i} \cdot 10^6}{N_c} \right) \right). \quad (1)$$

The factor 10^6 is required because $x_{c,i}$ is in USD millions while N_c is a raw population count. At low funding levels, the response is approximately linear with slope $B_{c,i}\kappa_i \cdot 10^6/N_c$ (the true derivative at $x = 0$); as funding increases, the exponential term decays and the response saturates at $B_{c,i}$.

2.5 Objective Functions

The problem is bi-objective. Country-level net impact is defined as the sum of intervention-specific DALYs averted, augmented by the ARV–Supply synergy term, and reduced by the ARV shortage penalty:

$$\text{Impact}_c(X) = \sum_{i \in I} \text{DALY}_{c,i}(x_{c,i}) + \gamma_c x_{c,\text{ARV}} x_{c,\text{Supply}} - P \cdot s_c^{\text{short}}. \quad (2)$$

Objective 1 — Maximise aggregate DALYs averted.

$$\max f_1(X) = \sum_{c \in C} \text{Impact}_c(X). \quad (3)$$

The first objective favours allocations that maximise aggregate health impact across all included countries.

Objective 2 — Minimise cross-country inequity. Let $\text{Impact}_c^{\text{pc}}(X) = 10^5 \cdot \text{Impact}_c(X)/N_c$ denote net DALYs averted per 100,000 PLHIV in country c . The inequity objective is defined as the standard deviation of $\text{Impact}_c^{\text{pc}}$ across countries with $N_c > 0$:

$$\min f_2(X) = \text{std}_{c:N_c>0}(\text{Impact}_c^{\text{pc}}(X)). \quad (4)$$

A lower value indicates that per-capita net DALY impact is more evenly distributed across countries. Normalising by population removes the trivial bias against small countries that would otherwise arise from an absolute worst-country objective.

The two objectives are in tension: f_1 rewards concentrating funds where marginal per-dollar DALY returns are highest (typically medium-population, high-burden countries), while f_2 rewards spreading impact more evenly across the population.

2.6 Constraints

Budget constraint

$$\sum_{c \in C} \sum_{i \in I} x_{c,i} \leq 200. \quad (5)$$

The total sum of money allocated to all countries cannot be more than USD 200M.

Absorption capacity.

$$0 \leq x_{c,i} \leq U_{c,i}, \quad \forall c \in C, \forall i \in I. \quad (6)$$

Each (country, intervention) pair has a maximum absorbable funding level beyond which additional funds yield no marginal benefit, reflecting realistic market saturation and population capacity.

Workforce–supply interdependency.

$$x_{c,\text{Workforce}} \leq \alpha \cdot (x_{c,\text{Supply}} + 1), \quad \forall c \in C. \quad (7)$$

Trained healthcare workers cannot be productively deployed beyond the supporting supply-chain infrastructure available. The additive constant of USD 1M represents residual pre-existing supply-chain infrastructure post-PEPFAR-cut, from which modest workforce deployment is possible even under zero new supply spending.

3 Data and Preprocessing

3.1 Data Sources

- **UNAIDS AIDSinfo Databook 2025** using the 2024 reporting year, the latest complete year in the source, for country-level PLHIV counts N_c , drawn from the “HIV2025Estimates_ByYear” sheet.
- **IHME Global Burden of Disease (GBD) 2023 HIV/AIDS estimates** for the maximum DALY-averting potential matrix $B_{c,i}$, with the four intervention columns already pre-split according to the 65/15/15/5 operational-dependency weights.
- **GHSC-PSM Product e-Catalog (2026)** for per-patient TLD contract pricing, anchoring both T_c and $U_{c,ARV}$.

3.2 Preprocessing Pipeline

1. Filter the UNAIDS dataset to 2024 (the latest complete year in the source).
2. Restrict to the 46 target countries listed in Section 2.1; sort into the canonical order used across all data structures using `pd.Categorical` with an ordered category list, ensuring row c refers to the same country in every matrix.
3. Parse UNAIDS PLHIV values, which are stored in mixed formats within column AE of the source sheets.
4. Calculate and load the IHME B-matrix and align to the same country ordering.
5. Compute T_c , $U_{c,i}$, $\beta_{c,i}$, and γ_c from the raw data using the formulas in Section 2.3.

3.3 Data Availability Limitation

Two of the 46 countries — Equatorial Guinea and São Tomé and Príncipe — have suppressed PLHIV values in UNAIDS 2024. Our pipeline conservatively maps these to $N_c = 0$, which propagates through $U_{c,i} = 0$ and forces zero allocation in every method. The optimisation therefore effectively covers 44 non-degenerate countries. Countries with small but reported PLHIV values (e.g. Cape Verde 3,900; Comoros 200) receive appropriately small but non-zero allocations.

Aggregating parsed values gives a total of 26,238,700 PLHIV captured across the 46 countries and a total ARV threshold of USD 872M — more than four times the available budget. This structural underfunding is a defining feature of the results (Section 7).

4 Reduced-Instance Exact Benchmark: Gurobi

The full formulation is nonlinear (via the saturating response, Eq. 1), non-convex (via the bilinear synergy $\gamma_c x_{c,ARV} x_{c,Supply}$), and bi-objective, and therefore not amenable to direct convex optimisation at the 46-country scale. To provide an exact-solver reference point on a tractable subproblem, we formulate a reduced linearised instance and solve it with Gurobi 12.0.

4.1 Reduced-Instance Formulation

- The 5 highest-burden countries by total $B_{c,\cdot}$, South Africa, Nigeria, Mozambique, Tanzania, Kenya, were used.
- The saturating response was replaced by the capacity-normalised linear approximation used in the code: $\text{DALY}_{c,i}(x_{c,i}) \approx \beta_{c,i} x_{c,i}$, treating $\beta_{c,i}$ as a constant per-dollar impact rate up to the absorption ceiling.
- The budget was scaled proportionally to the 5 countries' share of continental PLHIV, giving a reduced-instance budget of USD 117.4M.
- The bi-objective was scalarised using a weighted sum $F_\lambda = \lambda \hat{f}_1 + (1 - \lambda) z - 0.1 \cdot \text{shortage_rate}$, where $\hat{f}_1$ is the total DALY impact normalised by 5-country total burden, z is the max-min equity variable, and the shortage-rate penalty is added to discourage large ARV shortfalls. Higher λ prioritises aggregate DALYs; lower λ prioritises max-min equity across the five countries. We sweep $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$.
- The budget, absorption, and workforce-supply constraints from Section 2.6 were implemented.

4.2 Results

Table 1 reports the exact-optimal objective and country-level ARV allocation for each λ . Non-ARV allocations are zero everywhere except at $\lambda = 1$, where Nigeria additionally receives USD 4.8M for Supply Chain and USD 2.9M for Workforce (its ARV capacity is fully saturated at USD 66.48M, so residual budget flows into complementary interventions).

λ	Objective	S. Africa	Nigeria	Mozambique	Tanzania	Kenya
0.00	0.0720	59.45	15.24	19.06	13.72	9.91
0.25	0.0720	59.45	15.24	19.06	13.72	9.91
0.50	0.0765	29.63	66.48	9.50	6.84	4.94
0.75	0.1164	29.34	66.98 [†]	9.40	6.77	4.89
1.00	0.1787	0	74.17 [*]	0	0	43.21

Table 1: Gurobi exact-optimal allocations by weight λ (values in USD M, total country budget shown).

[†]Nigeria at $\lambda = 0.75$ includes USD 0.5M workforce spending. ^{*}Nigeria at $\lambda = 1$ additionally receives USD 4.8M (Supply) and USD 2.9M (Workforce) once its ARV capacity is saturated.

The optimal allocation shifts qualitatively with λ . At $\lambda \leq 0.25$ (equity-heavy), the budget is spread across all five countries with South Africa receiving the largest share (USD 59.4M), reflecting its dominant PLHIV count among the five and the max-min objective's push to raise every country's per-burden achievement. At $\lambda = 0.5$ – 0.75 , Nigeria's ARV capacity of USD 66.48M is fully saturated, drawing budget away from South Africa. At $\lambda = 1$, the solution collapses to a sparse two-country allocation: Nigeria and Kenya receive all funding, while South Africa, Mozambique, and Tanzania are zeroed out entirely.

This sparse corner-point behaviour is characteristic of LP relaxations: the linearised objective is a hyperplane in decision space, so its optimum lies at a vertex of the feasible polytope. Under linearisation, marginal DALY-per-dollar becomes the constant $\beta_{c,i} = B_{c,i}/U_{c,i}$, so the LP concentrates funds on the countries with the highest $\beta_{c,\text{ARV}}$ (Nigeria at $\approx 46,900$ DALYs/USD M and Kenya at $\approx 26,100$ DALYs/USD M among the five). The full 46-country problem, by contrast, is nonlinear in x through the concave saturating response (Eq. 1), which softens marginal returns as allocations grow and rewards diversification.

Scale caveat. The Gurobi objective values (0.07–0.18) are not directly comparable to the meta-heuristic aggregate DALY numbers (millions). The Gurobi objective is normalised (total impact divided by total 5-country burden; equity measured as a max-min achievement fraction) and operates on a reduced 5-country instance with a scaled budget. Additionally, the Gurobi benchmark uses a simplified max-min equity variable z rather than the standard-deviation inequity objective used in the main NSGA-II formulation (Eq. 4), because z admits a straightforward epigraph reformulation while standard deviation does not fit an LP directly. The Gurobi benchmark is therefore a small-instance sanity check on allocation patterns and LP-optimal behaviour under linearisation, not a target the full-instance metaheuristics should reach.

5 Main Metaheuristic: NSGA-II

The main optimisation method used is the Elitist Non-Dominated Sorting Genetic Algorithm II (NSGA-II). NSGA-II is suitable for this problem because the funding allocation is naturally bi-objective and there is no principled way to select a single fixed trade-off between aggregate DALYs and equity in advance. Rather than collapsing the two objectives into one scalar, NSGA-II maintains a population of candidate allocations and evolves a set of non-dominated solutions that approximate the Pareto frontier in a single run.

5.1 Solution Encoding

Each candidate solution is encoded as a 46×4 allocation matrix X (Section 2.2), with each entry storing the funding allocated to that country-intervention pair in USD millions. This encoding is natural because the budget, absorption capacity, and workforce–supply coupling constraints can all be checked directly on the matrix.

5.2 Initial Population Generation

To balance solution quality and diversity, the initial population of 60 chromosomes is generated using a mixed strategy:

- **Iterative greedy seed (1 chromosome).** The full standalone greedy heuristic (Section 6.1) is run first, and its final allocation is inserted verbatim into the initial population. This ensures that the DALY-maximising endpoint of the NSGA-II front matches or exceeds the greedy heuristic from generation 1, so all subsequent evolution can concentrate on expanding the frontier into more equitable regions.
- **Impact-ranked greedy seed (1 chromosome).** A simpler greedy variant that ranks country-intervention cells by $\beta_{c,i}$ and fills capacity in order. This provides a second impact-oriented seed with a different structure.
- **Needs-based proportional seed (1 chromosome).** Funding is allocated proportionally to country-level PLHIV, and within each country distributed across interventions using the 65/15/15/5 operational-dependency weights. This provides an equity-oriented seed.
- **Feasible random seeds (57 chromosomes).** Random allocations respecting the total budget and absorption capacity limits, generated by drawing weights uniformly and rescaling to the remaining budget. This preserves genetic diversity and explores the broader solution space.

5.3 Objective Evaluation and Non-Dominated Sorting

Each chromosome is evaluated on the two objectives (Eqs. 3, 4). Both objectives are internally negated where necessary so that the sorting routine minimises both. A solution p dominates q if p is no worse than q on both objectives and strictly better on at least one. The first Pareto front is the set of solutions not dominated by any other in the population; subsequent fronts are computed by removing the first front and repeating.

Within each front, the crowding distance

$$CD_i = \sum_{m=1}^M \frac{f_m^{(i+1)} - f_m^{(i-1)}}{f_m^{\max} - f_m^{\min}} \quad (8)$$

is computed to preserve solution diversity along the front. Solutions with larger crowding distance are preferred as they occupy sparser regions of the objective space.

5.4 Crossover, Mutation, and Repair

Arithmetic crossover. Two parents are combined using a per-country random mixing coefficient $\alpha \in [0, 1]$ ⁴⁶. The crossover is illustrated in the diagram below.

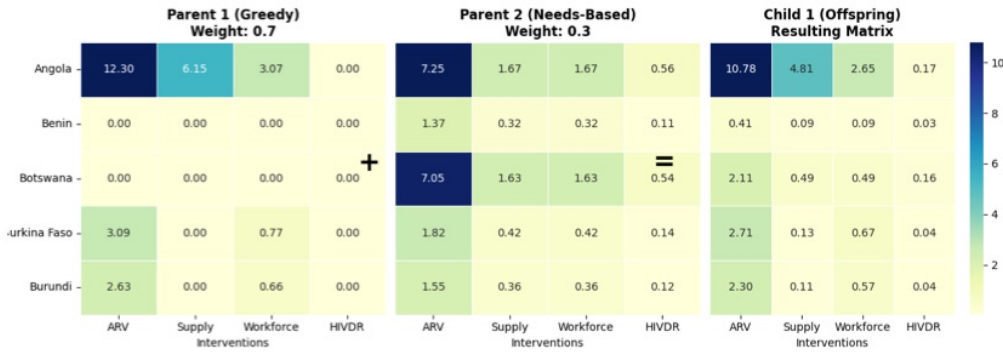


Figure 1: Illustration of arithmetic crossover.

Because the same α is used per country row, offspring inherit whole country-level allocation patterns from both parents while still exploring interpolated allocations. Crossover is applied with probability 0.9.

Standard single-point crossover treats decision variables independently and would arbitrarily inject or delete capital, immediately violating the global budget constraint. The arithmetic form preserves the budget under the (subsequent) repair step and is well-suited to continuous allocation problems.

Budget-shift mutation. Mutation is applied with probability 0.25 per chromosome. When triggered, the operator selects one donor cell uniformly at random from the set of positively-funded cells $\{(c, i) : x_{c,i} > 0\}$, and one receiver cell uniformly at random over all 46×4 cells. A random fraction of the donor's current allocation, drawn from $\mathcal{U}(0.05, 0.25)$, is transferred from donor to receiver. This preserves the global budget at the point of mutation (subject to the subsequent repair step) and produces small, local perturbations that allow the algorithm to explore refinements of promising allocations. A Gaussian-noise mutation was also considered but was found to violate the budget constraint too often and require heavier repair.

Repair. After crossover and mutation, offspring are repaired to enforce feasibility:

1. Clip negative allocations to zero.
2. Clip each allocation to $U_{c,i}$.
3. Enforce the workforce–supply constraint $x_{c,\text{Workforce}} \leq \alpha(x_{c,\text{Supply}} + 1)$.
4. If total spend exceeds USD 200M, scale allocations proportionally to the budget.
5. If total spend is below the budget, fill the remaining budget across cells with available capacity, weighted randomly.

5.5 Elitism and Survivor Selection

The parent population and offspring are merged into a combined pool of size $2N$. The pool is sorted into Pareto fronts, and fronts are accepted sequentially until the next front would overflow the population capacity N . The overflow front is admitted partially, prioritising solutions with the largest crowding distance to preserve diversity. This is the standard NSGA-II survivor-selection step and ensures elitist preservation of non-dominated solutions across generations.

5.6 Termination

The algorithm terminates when either the maximum generation count (300) is reached or when both the best aggregate-DALY value and the best inequity value fail to improve by more than $\epsilon = 10^{-4}$ (relative) for 40 consecutive generations. In our experiments, early stopping is not triggered before the maximum generation count, but convergence plots (Section 7.3) indicate that the front is effectively converged well before generation 300.

5.7 Hyperparameter Tuning

Table 2 summarises the tuned hyperparameter values used in the final experiments.

Parameter	Value
Population size	60
Maximum generations	300
Early-stopping patience	40 generations
Early-stopping tolerance	10^{-4}
Crossover rate	0.9
Mutation rate (per chromosome)	0.25
Maximum mutation transfer fraction	0.25
Tournament selection size	2

Table 2: NSGA-II hyperparameters.

The convergence trace in Section 7.3 indicates that the front is effectively converged by roughly generation 100. The 300-generation ceiling was retained as a conservative bound to demonstrate full convergence.

The NSGA-II hyperparameters were chosen through pilot runs that balanced solution quality and runtime. A population size of 60 was large enough to maintain a diverse Pareto front while keeping non-dominated sorting computationally manageable. The generation limit of 300 was retained as a conservative upper bound, although convergence diagnostics later showed that most hypervolume

improvement occurred within the first 100 generations. A crossover rate of 0.9 was used to encourage recombination of high-quality country-level allocation patterns, while a mutation rate of 0.25 was used to introduce budget-shift perturbations without destroying feasible structure. The mutation transfer fraction was capped at 0.25 so that each mutation remained a local reallocation rather than a complete restructuring of the solution. Early-stopping patience of 40 generations and tolerance of 10^{-4} were selected to avoid stopping during short plateaus in multi-objective search.

6 Baseline and Alternative Methods

6.1 Baseline: Greedy Marginal-DALY Heuristic

A greedy marginal-DALY heuristic is implemented as a simple impact-focused baseline. Starting from a zero allocation, the heuristic iteratively evaluates the marginal net DALY gain of adding a USD 1M block to each feasible country-intervention cell and assigns the block to the cell with the largest gain. The process continues until the budget is exhausted, subject to the capacity and workforce-supply constraints.

This method is intuitive and computationally cheap, but myopic: it does not consider how current decisions affect future allocation flexibility or equity. It is expected to perform strongly on aggregate DALYs but poorly on inequity, providing a natural point of comparison against NSGA-II's Pareto front.

6.2 Alternative Metaheuristic: Simulated Annealing

Simulated annealing (SA) with weighted-sum scalarisation is implemented as an alternative metaheuristic benchmark. Unlike NSGA-II, SA operates on a single solution and requires the bi-objective problem to be scalarised. Let $\hat{f}_1$ and $\hat{f}_2$ denote the normalised objectives (rescaled to $[0, 1]$ using reference ranges anchored on the greedy and needs-based seeds). For a trade-off weight $\lambda \in [0, 1]$:

$$F_\lambda(X) = \lambda \hat{f}_1(X) + (1 - \lambda) \hat{f}_2(X). \quad (9)$$

Higher λ prioritises aggregate DALYs; lower λ prioritises equity.

A neighbouring solution is generated by transferring a random fraction of budget between two country-intervention cells (the same budget-shift operator as in NSGA-II mutation). If the scalar objective improves, the neighbour is accepted; otherwise it is accepted with probability $\exp(-\Delta/T)$, where T is the current temperature. Temperature decays geometrically: $T_{t+1} = c \cdot T_t$ with $c = 0.9997$ (giving $T_{20,000} \approx 0.0025T_0$).

For a like-for-like comparison to NSGA-II, SA is warm-started from the seed most aligned with the current λ (greedy for $\lambda \geq 0.5$, needs-based for $\lambda < 0.5$), and run for 20,000 iterations across $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$.

Because SA returns one scalar-optimal solution per λ rather than a full frontier, five separate runs are required to approximate the trade-off structure that NSGA-II produces in a single run.

7 Performance Benchmark

We compare NSGA-II against the greedy baseline and SA at all five weights on the full 46-country instance with USD 200M budget and identical objective evaluations across methods. Because the problem is bi-objective, we report both aggregate impact (net DALYs averted) and equity (inequity score, in f_2 units).

7.1 Comparison Summary

Table 3 summarises the best solution returned by each method under a common random seed.

Method	Net DALYs Averted	Inequity Score	ARV Shortfall (USD M)
Greedy Marginal DALY	4,298,745	89,949	708
Simulated Annealing ($\lambda = 0$)	1,998,629	60,164	818
Simulated Annealing ($\lambda = 0.25$)	2,362,897	64,941	740
Simulated Annealing ($\lambda = 0.5$)	4,025,437	110,552	693
Simulated Annealing ($\lambda = 0.75$)	4,071,999	125,507	690
Simulated Annealing ($\lambda = 1$)	4,100,415	128,745	687
NSGA-II — best-DALYs endpoint	4,313,203	100,700	709
NSGA-II — knee point	4,240,734	60,159	710
NSGA-II — best-equity endpoint	3,956,995	47,634	713

Table 3: Best solution returned by each method. All allocations use the full USD 200M budget.

The NSGA-II knee point provides a substantially better balanced allocation than greedy: it preserves 98.6% of greedy’s aggregate DALYs (4.24M vs 4.30M) while reducing inequity by 33% (60,159 vs 89,949). Since greedy has higher aggregate DALYs but the knee has lower inequity, the two solutions are mutually non-dominated — the knee does not strictly Pareto-dominate greedy — but it offers a more attractive efficiency–equity trade-off for a decision-maker who weighs both objectives. Neither greedy nor any SA run reaches this combination: greedy has poorer equity, and SA runs at comparable inequity ($\lambda \leq 0.25$) achieve less than half of NSGA-II’s aggregate impact. The NSGA-II best-DALY endpoint at 4,313,203 DALYs also slightly exceeds greedy’s 4,298,745, indicating that population-based search discovered a marginally better impact-only allocation than the myopic greedy construction, primarily through mutation-driven refinement of greedy’s initial solution.

Simulated annealing under weighted-sum scalarisation performs sensibly across weights: $\lambda = 0$ concentrates on equity (60,164 inequity, 2.00M DALYs), $\lambda = 1$ concentrates on impact (4.10M DALYs, 128,745 inequity), and intermediate weights interpolate. However, every SA solution is Pareto-dominated by some point on the NSGA-II frontier. The SA $\lambda = 0$ solution has essentially the same inequity as the NSGA-II knee point but 53% fewer DALYs. This confirms two structural weaknesses of scalarised SA in this bi-objective setting: (a) each weight requires a separate run to expose a single trade-off point, whereas NSGA-II produces the entire frontier in one pass, and (b) the random-shift neighbourhood operator explores the 184-dimensional continuous space less efficiently than crossover-driven exploration across a population.

7.2 Pareto Frontier Analysis

Figure 2 plots the final Pareto front. The frontier takes a characteristic concave shape: net DALYs decline gradually as inequity is reduced from $\approx 100\text{k}$ toward $\approx 55\text{k}$, then more steeply toward the equity-maximising endpoint. The **knee point** — selected as the frontier solution closest to the utopia point (max DALYs, min inequity) after normalising both objectives to $[0, 1]$ — sits at (4,240,734 DALYs, 60,159 inequity). Moving along the frontier from the knee:

- Toward higher DALYs: a 1.7% gain in aggregate impact (4.24M $\rightarrow$ 4.31M) costs a 67% increase in inequity (60.2k $\rightarrow$ 100.7k).
- Toward lower inequity: a 21% reduction in inequity (60.2k $\rightarrow$ 47.6k) costs a 6.7% reduction in aggregate DALYs (4.24M $\rightarrow$ 3.96M).

Both directions face diminishing returns, but the marginal cost is far steeper on the DALY-maximising side. The knee is a real inflection in the trade-off, not a plotting artefact.

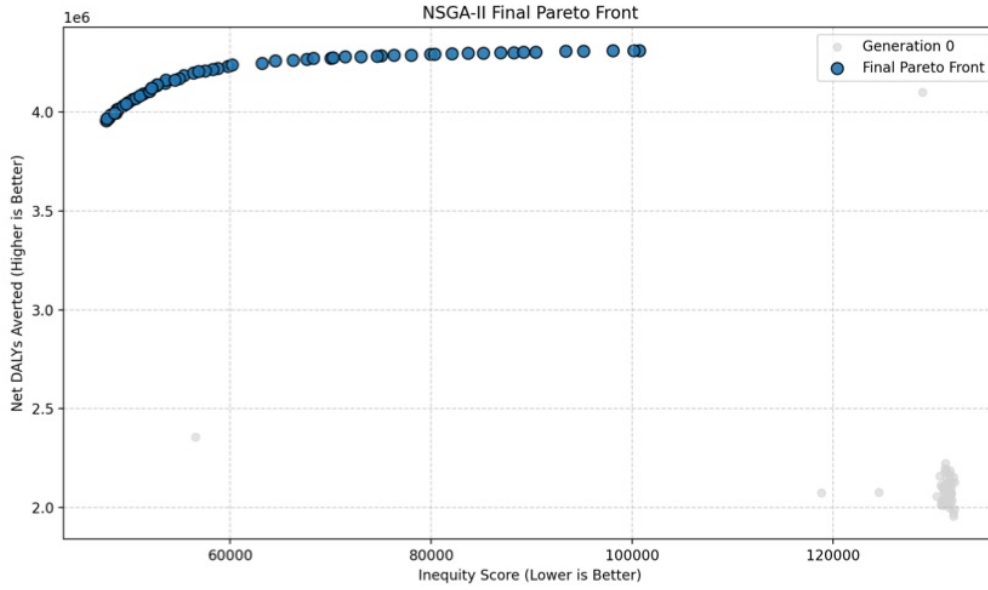
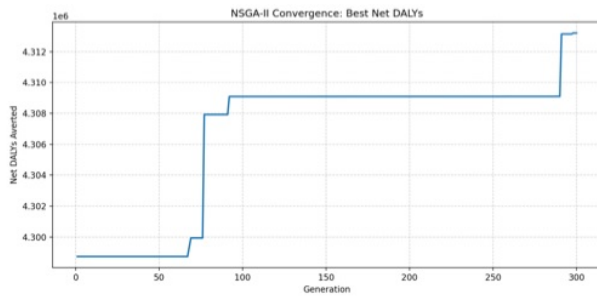
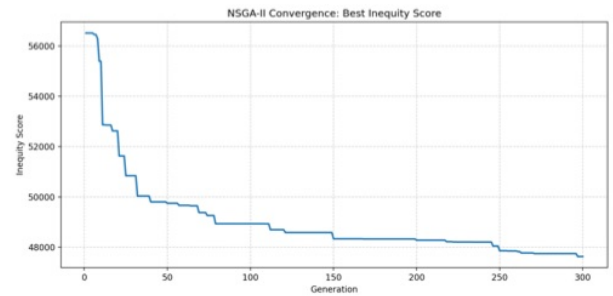


Figure 2: Final NSGA-II Pareto front (blue) after 300 generations, overlaid on the initial random population (grey). The front contains 60 non-dominated solutions spanning DALYs from 3.96M to 4.31M and inequity from 47,634 to 100,700.

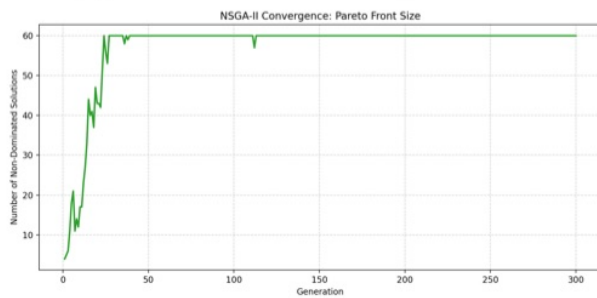
7.3 Convergence Behaviour



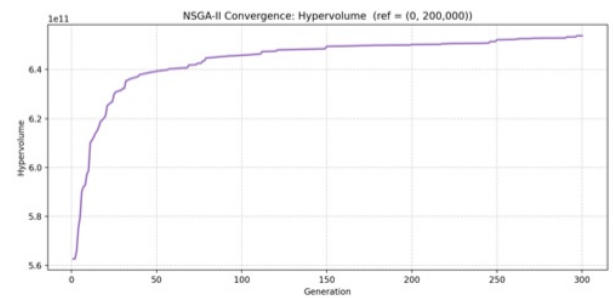
(a) Best net DALYs across generations.



(b) Best inequity score across generations.



(c) Size of the non-dominated front.



(d) 2D hypervolume (reference (0, 200,000)).

Figure 3: NSGA-II convergence over 300 generations.

Four complementary metrics characterise NSGA-II convergence (Figure 3). First, the best inequity score decreases from approximately 56,500 at generation 1 to 47,634 at termination, with the largest improvements occurring in the first 40 generations and smaller refinements thereafter. Second, the best net DALYs improve only slightly, from 4,298,745 at generation 1 to 4,313,203, because the standalone greedy solution is injected as an initial seed and already provides a strong impact-focused allocation. Third, the number of non-dominated solutions grows from 4 at generation 1 to the full

population size of 60 by around generation 30, indicating that the population rapidly spreads across the trade-off frontier. Finally, the 2D hypervolume increases from 5.63×10^{11} to 6.55×10^{11} , a 16.4% improvement, with most gains captured within the first 100 generations.

The convergence results show that NSGA-II is not mainly improving the maximum-DALY endpoint, since that endpoint is already strong due to greedy seeding. Instead, its main contribution is expanding and refining the Pareto frontier, especially by finding lower-inequity allocations that preserve high aggregate DALYs. The hypervolume and front-size plots indicate that the search is effectively converged by generation ~ 100 . Additional generations from 100 to 300 yield only marginal refinements, suggesting that 100–150 generations would be sufficient for practical use.

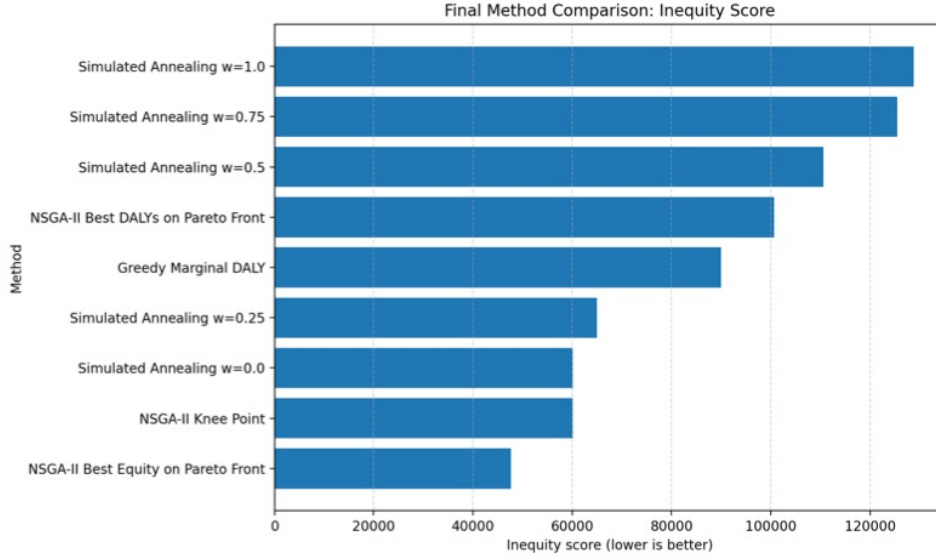


Figure 4: Final inequity comparison across methods. Lower values indicate a more balanced distribution of per-capita net DALY impact across countries. NSGA-II achieves the lowest-inequity endpoint overall, while the NSGA-II knee point attains inequity comparable to equity-focused simulated annealing but with substantially higher aggregate DALYs.

Figure 4 highlights the advantage of NSGA-II on the equity dimension. Although the greedy heuristic and DALY-focused simulated annealing achieve strong aggregate impact, they produce substantially higher dispersion in per-capita net DALYs across countries. The NSGA-II best-equity endpoint achieves the lowest inequity score overall, while the NSGA-II knee point obtains almost the same inequity as the equity-focused SA run but with far higher aggregate DALYs. This supports the use of NSGA-II as the main method, since it identifies trade-off solutions that single-objective or scalarised methods do not capture.

7.4 Runtime and Time Complexity

Table 4 reports runtimes of each method on a mid-range consumer laptop (Intel i5-class CPU, single-threaded Python 3.11 with NumPy 2.x), together with their asymptotic time complexity. Runtimes are approximate ($\pm 20\%$) and vary with hardware, but the relative ordering is stable across machines.

Per-method complexity. Greedy performs $K = 200$ increments, each evaluating the marginal DALY gain over all (c, i) cells and selecting the best: $\mathcal{O}(K \cdot |C| \cdot |I|) \approx 3.7 \times 10^4$ marginal evaluations, completing in seconds. SA performs $T = 20,000$ iterations per weight, each involving one neighbour generation, one objective evaluation, and one repair (both $\mathcal{O}(|C| \cdot |I|)$), giving $\mathcal{O}(3.7 \times 10^6)$ operations per weight and five weights linearly. NSGA-II is dominated by fast non-dominated sorting at $\mathcal{O}(M^2)$ per generation (following Deb et al. 2002), multiplied by the per-evaluation cost, giving $\mathcal{O}(G \cdot M^2 \cdot$

Method	Approx. Runtime	Time Complexity (per problem instance)
Greedy Marginal DALY	~ 8 s	$\mathcal{O}(K \cdot C \cdot I)$
Simulated Annealing (all 5 weights)	~ 4 – 5 min	$\mathcal{O}(T \cdot C \cdot I)$ per weight
NSGA-II (60 pop, 300 gen)	~ 3 – 4 min	$\mathcal{O}(G \cdot M^2 \cdot C \cdot I)$
Gurobi (5-country reduced LP, 5 λ)	~ 3 s	polynomial in problem size (LP)
Full pipeline (all four methods)	~ 10 min	—

Table 4: Approximate wall-clock runtime and asymptotic time complexity of each method. $|C| = 46$ countries, $|I| = 4$ interventions, $K =$ number of USD 1M budget increments ($=200$), $T =$ SA iterations per weight ($=20,000$), $G =$ NSGA-II generations ($=300$), $M =$ NSGA-II population size ($=60$).

$|C| \cdot |I| \approx 2 \times 10^8$ operations. Gurobi solves 20 continuous variables plus slacks per λ , well within the polynomial-time LP regime and empirically sub-second.

Observations. NSGA-II is roughly $30\times$ slower than greedy but produces 60 non-dominated solutions rather than one, so per-solution cost is comparable. NSGA-II is slightly faster than the full SA sweep and produces the entire frontier in one run rather than five scalarised solutions. The Gurobi LP is the fastest exact option, but only on the linearised reduced instance; the full 46-country nonlinear problem would require non-convex solvers that scale much less favourably.

8 Insights from the Optimal Solution

8.1 The Scenario is Structurally Underfunded

The most important highlight is not any particular allocation pattern but the underlying supply–demand gap by design. Summing T_c across the 46 countries gives a total ARV need of USD 872M; more than four times the available USD 200M budget. Even the best NSGA-II solution leaves an ARV shortfall of USD 709M; the knee point leaves USD 710M. Across the main balanced solutions — greedy, the NSGA-II frontier, and the impact-weighted SA runs — roughly 79–82% of the calculated minimum ARV need remains unmet, while the equity-weighted SA runs leave still larger shortfalls. This budget level is used to simulate a severe scarcity scenario in which a single philanthropic donor cannot fully replace large-scale public funding. The USD 200M budget is therefore not intended to close the full continental ARV gap, but to create a realistic high-conflict allocation setting where trade-offs between efficiency and equity become visible.

Under this constraint, the optimisation problem is no longer “how to fund all interventions optimally” but rather “where to concentrate the scarce ARV supply”. Any conclusions about optimal supply chain, workforce, or HIVDR spending are conditional on the ARV threshold being modelled as a soft-penalty rather than a hard cutoff. Under a substantially larger budget (approximately USD 800M and above, approaching the total ARV threshold), the balance across intervention types would shift qualitatively as the most severe ARV scarcity is relaxed. However, under the current implementation, the ARV–Supply synergy term itself remains a small proxy term; a stronger empirical synergy model would require calibrated ART coverage or supply-chain coverage data.

8.2 ARV Dominates Every Allocation

Across all methods, ARV procurement absorbs 82–87% of the total budget. In the NSGA-II knee-point solution, ARV receives $\sim$ USD 173M, with the remainder split across Supply, Workforce, and HIVDR. Greedy is even more concentrated: ARV USD 164.0M, Supply USD 21.0M, Workforce USD 15.0M, HIVDR USD 0.

This concentration is driven by the shortage penalty structure. With $P = 1000$ DALYs per USD M of shortfall and total shortfall over USD 700M, the marginal DALY return from any additional ARV dollar is substantially higher than from Supply, Workforce, or HIVDR. The ARV–Supply synergy term $\gamma_c x_{\text{ARV}} x_{\text{Supply}}$ is not large enough to displace this pattern under the USD 200M budget. This accurately reflects that under a binding drug supply shortage, non-drug interventions have low immediate marginal impact.

8.3 Allocation Follows Marginal Per-Dollar Return rather than Absolute Burden

A naive expectation is that funding should scale with total HIV burden: South Africa, with 7.8M PLHIV and the highest total DALY potential in the dataset ($B_{c,\text{total}} \approx 6.0\text{M}$), should receive the largest share. In practice, greedy allocates USD 0 to South Africa and USD 75M to Nigeria (2.0M PLHIV, $B_{c,\text{total}} \approx 4.8\text{M}$).

This apparent paradox is a direct consequence of the saturating response function (Eq. 1). Marginal DALY-per-dollar scales with the dollars-per-person ratio x/N : USD 1M produces roughly $B_{c,\text{ARV}} \cdot \kappa_{\text{ARV}} \cdot 10^6 / N_c$ marginal DALYs. For representative countries:

Country	PLHIV	$B_{c,\text{ARV}}$	$\kappa_{\text{ARV}} \cdot 10^6 / N_c$	Marginal DALYs / USD M
Cameroon	510,000	844,000	0.039	~33,000
Nigeria	2,000,000	3,117,000	0.010	~31,000
Ethiopia	600,000	841,000	0.033	~28,000
South Africa	7,800,000	3,903,000	0.003	~10,000

South Africa has the highest absolute burden, but each USD 1M covers a much smaller fraction of its at-risk population, so the marginal per-dollar return is roughly a third of Nigeria’s. Cameroon, Ethiopia, and Nigeria all receive substantial allocations for this reason. NSGA-II shifts some budget toward South Africa relative to greedy: greedy allocates it USD 0, while the NSGA-II knee point allocates USD 1.48M. This occurs because the equity objective penalises the extreme concentration produced by greedy, but the adjustment remains small in absolute terms relative to South Africa’s burden, a pattern examined further in Section 10.3.

8.4 NSGA-II Improves the Efficiency–Equity Trade-off Over Greedy

At the knee-point solution, NSGA-II delivers a strong efficiency–equity improvement over the standalone greedy heuristic: 98.6% of greedy’s aggregate DALYs at 33% lower inequity. The two solutions are mutually non-dominated in the strict Pareto sense (greedy has higher DALYs; the knee has lower inequity), but the knee is a substantially more attractive point for any decision-maker who weighs both objectives. Greedy did not discover this solution because its single-objective myopic construction cannot express the equity dimension.

The mechanism is straightforward. Greedy allocates funds to whichever cell offers the largest immediate marginal DALY gain at each step, which is locally optimal for aggregate impact in isolation, but concentrates funding in a small number of countries and leaves smaller countries essentially unfunded. NSGA-II’s population-based search evaluates many redistributions of the same total budget simultaneously and, through crossover and mutation, discovers small reallocations from over-funded high-burden countries to under-funded medium-burden countries that improve equity substantially without materially harming aggregate impact.

At the DALY-maximising endpoint, NSGA-II also marginally exceeds greedy (4.313M vs 4.299M DALYs), obtained through mutation-driven exploration of allocations near greedy’s construction. This demonstrates that the metaheuristic search adds value even on the objective where greedy is expected to be strongest.

8.5 HIVDR Underfunding is a Model Artefact, Not a Policy Recommendation

Every method allocates near-zero funding to HIVDR surveillance. This should not be interpreted as a recommendation. Our static single-period model captures only the immediate marginal DALY benefit of HIVDR, which is small under the 5% operational-dependency weight. In reality, HIVDR's value is dynamic: surveillance protects the future effectiveness of the first-line ARV regimen by detecting resistance mutations early and enabling regimen switching before treatment failure cascades and drives transmission of resistant strains. A dynamic multi-period model with regimen-switching cost differentials between first- and second-line therapy would likely allocate substantially more to HIVDR.

8.6 Countries with Suppressed Data Receive Zero

Equatorial Guinea and São Tomé and Príncipe are mapped to $N_c = 0$ by our conservative pre-processing (Section 3.3) and consequently receive zero allocation in every method. This is a hard consequence of the input data, not a modelling choice. The optimisation therefore effectively covers 44 non-degenerate countries. Future work should apply regional-average imputation to include the two suppressed countries.

9 Assumptions

The following simplifying assumptions underpin our model. Each is a deliberate scope decision; the effect of relaxing each is discussed briefly.

- **Static single-period horizon.** The model allocates the USD 200M budget once, with impact measured as DALYs averted over a nominal five-year window. Dynamic effects (regimen switching triggered by resistance, HIV transmission dynamics, epidemic trajectory changes, inter-year budget carryover) are not simulated.
- **Deterministic epidemiological parameters.** The IHME B-values, UNAIDS PLHIV estimates, and derived β , κ , T , U parameters are treated as point estimates without uncertainty bands. Sensitivity to parameter uncertainty (e.g., via Latin hypercube sampling) is left for future work.
- **Separable intervention effects with one synergy term.** The saturating response assumes intervention impacts add independently at the country level, augmented by a single ARV–Supply bonus. Real-world intervention interactions may be more complex (e.g., workforce–HIVDR complementarity for early resistance detection).
- **Heuristic split of country DALY burden across interventions.** The 65/15/15/5% weights are based on operational-dependency reasoning rather than empirical decomposition. Actual DALY attribution across intervention line-items is programme- and context-specific.
- **Burden-normalised proxy for γ_c .** The ARV–Supply synergy coefficient is calibrated from a burden-share proxy rather than the ART coverage gap intended in the proposal, because per-country ART coverage data was not present in the assembled dataset.
- **Uniform per-patient cost assumptions.** Drug cost (USD 33.24/patient/year), supply-chain markup (USD 16.62/patient/year), workforce cost (USD 8.31/patient/year), and HIVDR test cost (USD 20/sample) are treated as continent-wide constants. Real per-patient costs vary substantially with local wages, transport distances, and existing infrastructure.
- **Zero-capacity treatment of data-suppressed countries.** Equatorial Guinea and São Tomé and Príncipe are forced to zero allocation.

- **Baseline supply infrastructure represented by a USD 1M offset** in the workforce–supply coupling. This is a modelling choice, not a data-driven parameter.
- **No procurement lead times or in-year absorption limits.** All funding is assumed immediately deployable.
- **Continuous funding variables.** Procurement and workforce deployment happen in discrete units in practice, but the continuous relaxation is a reasonable approximation at aggregate-country level.

10 Limitations

The Pareto-optimal allocations reported in Sections 6–8 should not be read as direct policy recommendations. Beyond the scope decisions catalogued in Section 9, our model contains two structural properties that jointly determine which countries the optimum favours. Neither arises from a deliberate ethical judgement about how funding should be distributed; both are consequences of the mathematical form of the objective.

10.1 Marginal-Return Bias Favours Medium-Population Countries

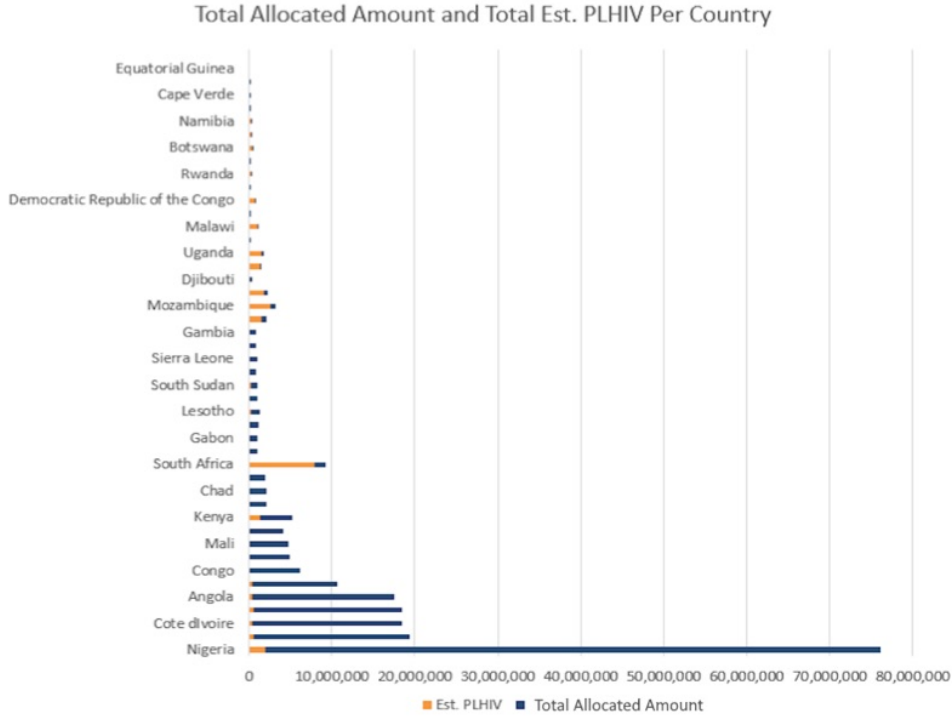


Figure 5: Combined Bar Chart of Total Allocation and Total Est. PLHIV per Country

The saturating impact response of Eq. 1 has marginal per-dollar DALY return $B_{c,i} \cdot \kappa_i \cdot 10^6 / N_c$ at low allocation levels the regime in which every solution operates, given a USD 200M budget against a USD 872M total threshold. This marginal return scales inversely with population, and the knee-point solution makes the resulting distortion stark: Nigeria, with an estimated 2.0M people living with HIV, receives USD 74.12M; 37% of the entire budget while South Africa, with an estimated 7.8M people living with HIV, nearly four times Nigeria’s burden, receives only USD 1.48M. USD 1M yields approximately 33,000 marginal DALYs in Cameroon (0.51M PLHIV) but only 10,000 in South Africa, because the same dollar covers a far larger fraction of a smaller country’s at-risk population. A greedy

or population-based optimiser therefore systematically funds smaller- and medium-population, high-per-capita-burden countries first, even when the largest country on the continent by absolute HIV burden is left funded at a level barely above a country one-hundredth its size. This is not a case of the model failing to notice South Africa’s need, rather it is the mathematically correct response to an objective function that rewards per-dollar efficiency rather than absolute burden.

10.2 The Absolute-Shortfall Penalty Does Not Counteract the Bias

The shortage penalty $P \cdot s_c^{\text{short}}$ is defined in absolute dollars of unmet threshold, and the marginal penalty relief from adding USD 1M of ARV to any country in shortfall is exactly $P = 1000$ DALYs — a constant that does not vary with country size. The penalty therefore does not directly discriminate at the margin. But equally, it does not correct for the marginal-return bias in §10.1: a larger country in shortfall receives no per-dollar advantage from the penalty relief that would offset its smaller per-dollar saturating return. In the current formulation the marginal DALY return dominates, and the penalty acts as a floor that only becomes binding once a country’s threshold is nearly met, rather than as an active pull toward large-population countries.

Alternative penalty formulations would embody different implicit fairness principles. A proportional penalty $P \cdot s_c^{\text{short}}/T_c$ would yield marginal relief $P/T_c \propto 1/N_c$, which would in fact worsen the bias toward small countries. A quadratic penalty $P \cdot (s_c^{\text{short}})^2$ would give marginal relief $2P \cdot s_c^{\text{short}}$, growing with the absolute shortfall, and would give large-population countries the pull toward funding that the current absolute-linear form does not provide. Neither formulation is correct in absolute terms; each is a choice about how the model expresses “priority.”

10.3 Whom the Model Serves

Combining the two effects above, our model behaves as a triage allocator: it funds where the next dollar goes furthest, not where the most people need help. The empirical pattern in the NSGA-II knee point reflects this precisely. **The model serves medium-population, high-per-capita-burden countries most strongly.** Nigeria (2.0M PLHIV) alone receives 30–38% of every solution; Cameroon, Ethiopia, Angola, and Côte d’Ivoire (each 0.5–0.7M PLHIV) receive substantial allocations. Both extremes are under-served: South Africa (7.8M PLHIV) receives near-zero funding despite carrying the largest absolute HIV burden in the dataset, and the smallest countries (Cape Verde, Comoros) receive only symbolic allocations because their absolute burden is genuinely small.

This pattern is internally consistent with a utilitarian objective (maximise aggregate DALYs) under a concave saturating response. It is defensible under strict cost-effectiveness reasoning — one additional dollar averts more DALYs in Cameroon than in South Africa — but it conflicts with several other ethical frameworks that a decision-maker might reasonably adopt:

- A **prioritarian** framework (“help the worst-off first”) would weight allocations toward high-burden countries regardless of per-dollar returns. Under this view South Africa’s exclusion is difficult to justify.
- A **rights-based** framework treating ART access as an entitlement would allocate proportionally to PLHIV counts, giving both South Africa and Nigeria substantially more even shares.
- A **population-adjusted max-min** framework would minimise the worst per-capita achievement rather than the standard deviation across countries, pulling further budget toward large under-served countries than our std-based inequity objective does.

Our inequity objective (Eq. 4) partially counteracts the marginal-return concentration by penalising uneven per-capita impact, and the equity endpoint of the Pareto front is meaningfully more distributed than the DALY-maximising endpoint. But even at that endpoint, South Africa receives

less than a burden-proportional share. The equity objective, as defined, does not fully close the gap between what the utilitarian formulation optimises and what several defensible ethical frameworks would prescribe.

10.4 Defensible Uses of the Model

Given these caveats, the model is most defensibly used as:

- **A structural diagnostic.** The core finding — that USD 200M cannot close a USD 872M ARV threshold under any allocation — is robust to the specific objective form and is a policy-relevant conclusion in its own right.
- **A trade-off explorer.** The Pareto frontier makes explicit the aggregate-DALY cost of moving toward more equitable per-capita distributions, and the knee point offers a defensible balanced compromise between the two objectives *as specified*.
- **A baseline for framework comparison.** Rerunning with a proportional-shortfall penalty, a quadratic penalty, a max-min equity objective, or a per-capita-burden reward term would show how sensitive the allocation is to modelling choices, and quantify how much of the observed pattern is driven by the objective versus the data.

The model is not defensibly used as an oracle that identifies which specific country should receive funding next. Any such recommendation requires an explicit ethical framework and a corresponding reformulation of the objective, not merely a rerun of the current model with different weights.

A one-at-a-time sensitivity analysis on P , γ_c , and total budget is provided in Appendix A.4. It shows that ARV concentration is a robust feature of the USD 200M scarcity scenario — not an artefact of a single calibrated constant — while relaxing the budget substantially changes country-level allocation patterns, including a roughly 37-fold increase in South Africa’s allocation at USD 800M.

11 Conclusion

This project formulated HIV intervention funding allocation across 46 African countries as a bi-objective non-convex nonlinear program: maximise aggregate DALYs averted while minimising cross-country inequity, subject to a USD 200M budget, per-country absorption capacity, and workforce–supply coupling. The bilinear ARV–Supply synergy term and the saturating response function preclude direct convex optimisation at the full 46-country scale, motivating the use of multi-objective metaheuristic methods.

We implemented NSGA-II as the main method, benchmarked against a greedy marginal-DALY heuristic and simulated annealing under weighted-sum scalarisation. NSGA-II with a greedy-seeded initial population, arithmetic crossover, budget-shift mutation, and standard non-dominated sorting produced a Pareto frontier of 60 non-dominated solutions after 300 generations. Convergence, measured by 2D hypervolume, was effectively complete by generation 100. The most defensible single allocation on the frontier is the knee-point solution, which achieves 4,240,734 DALYs averted (98.6% of the greedy heuristic’s aggregate impact) while reducing cross-country inequity by 33%. This is a strong efficiency–equity improvement over greedy — although not a strict Pareto improvement, since greedy has marginally higher aggregate DALYs — and represents a more attractive trade-off than any single-point SA weight can produce.

The most important finding is structural rather than algorithmic. The total minimum ARV threshold across the 46 countries is USD 872M — more than four times the available budget — so the optimisation problem is really “where to concentrate scarce ARV supply,” not “how to fund all interventions optimally.” This reframing explains most of the observed allocation patterns: ARV absorbs

82–87% of every solution, Nigeria alone receives 30–38% of the budget (driven by marginal per-dollar return under the saturating response, not by absolute HIV burden), and HIVDR surveillance receives near-zero funding despite its long-run importance.

At the knee point this pattern reaches its starkest expression: South Africa, carrying the largest absolute HIV burden of any of the 46 countries at an estimated 7.8M PLHIV, receives only USD 1.48M — less than two dozen other countries whose PLHIV populations are a fraction of its size — while Nigeria, at roughly a quarter of South Africa’s burden, receives USD 74.12M. This is not a defect unique to our formulation; it reproduces a documented tension in cost-per-DALY-driven global health philanthropy more broadly, where efficiency and absolute need are, by construction, not the same target.

This concentration is driven primarily by the shortage penalty structure. With $P = 1000$ DALYs per USD M of shortfall and total shortfall over USD 700M, the marginal penalty reduction from additional ARV funding is much larger than the additive ARV–Supply interaction term. In the implemented formulation, $\gamma_c x_{c,ARV} x_{c,Supply}$ is numerically small because x is measured in USD millions and γ_c is bounded between 0.001 and 0.010. Hence, the synergy term does not materially displace the ARV-dominant allocation pattern in any tested scenario.

Several extensions are natural. A dynamic multi-period model would capture the true value of HIVDR surveillance — which the static formulation systematically underweights — and allow inter-year budget carryover. Replacing the burden-normalised γ_c proxy with actual ART coverage data would strengthen the calibration of the ARV–Supply synergy term. Sensitivity analysis via Latin hypercube sampling would quantify how robust the Pareto frontier is to uncertainty in the IHME B values and UNAIDS PLHIV estimates. Regional imputation for countries with suppressed UNAIDS values would extend effective coverage from 44 to the full 46 countries. And, as discussed in Section 10.3, replacing the current absolute-shortfall penalty and std-based inequity objective with alternative fairness formulations (proportional shortfall, quadratic shortfall, or max-min per-capita achievement) would allow the model to serve as a comparative framework rather than a single-answer optimiser.

Beyond the model, the more fundamental takeaway for policy is that an efficient allocation cannot substitute for adequate absolute funding: no reallocation of USD 200M closes a USD 872M gap, and the honest role of an optimisation model in this setting is to distribute the shortfall as equitably as possible under an explicitly-chosen fairness framework, not to conjure impact from a budget that structurally cannot deliver it.

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A Appendix

A.1 Author Contributions

Name	Contribution
Ho Jun Lin	Data preprocessing pipeline for UNAIDS and IHME datasets, country filtering and data-quality checks. Formulation of the ARV threshold and absorption-capacity parameters. Implementation and execution of the Gurobi reduced-instance benchmark. Researched on ethical concerns and results interpretation with qualitative comparison. Contributed to benchmark interpretation and report writing.
Sng Zhenhao	Implementation of the NSGA-II pipeline including solution encoding, objective evaluation, non-dominated sorting, crowding distance, arithmetic crossover, budget-shift mutation, repair procedure, convergence tracking, and knee-point selection. Also implemented benchmark comparison logic, generated all plots and CSV outputs, and contributed to methodology and results writing.
Woon Zee Ning	Background research on HIV treatment, ARV supply disruption, HIVDR surveillance, intervention cost assumptions, and public-health motivation. Sourcing of datasets. Justification of parameter values, source citation and interpretation. Contributed to the ethical framing, limitations, and final report editing.

All modelling decisions, data processing, experiments, and final interpretations were reviewed and jointly agreed by the group.

A.2 Reproducibility

The submitted code file `hst_hiv_optimisation_full.py` reproduces all main experiments. Required inputs are `hiv_data.xlsx` (UNAIDS AIDInfo 2025) and `edited_IHME-GBD_2023_DATA-f55e6b49-1.xlsx` (IHME GBD 2023), placed alongside the script. On execution, the script writes the Pareto frontier, convergence plots, method-comparison summary, and per-country allocation CSVs into `outputs_hiv_optimisation/`. Required packages: `numpy`, `pandas`, `matplotlib`, `openpyxl`. The optional Gurobi benchmark additionally requires `gurobipy` and a valid licence (a free academic licence was used here). Installation and execution:

```
pip install numpy pandas matplotlib openpyxl gurobipy
python hst_hiv_optimisation_full.py
```

The runtime constants `NSGA_MAX_GENERATIONS`, `NSGA_PATIENCE`, and `SA_ITERATIONS` are exposed near the top of the script; default values reproduce Sections 6–7.

A.3 Use of AI Assistance

Large language models (specifically Anthropic’s Claude) were used as writing and debugging assistants: to refine phrasing, identify consistency issues between the implementation and report, verify mathematical claims (including correction of a false Pareto-dominance claim in an earlier draft), and improve code documentation. All modelling decisions, data processing, algorithm implementations, experimental runs, and final interpretations were reviewed and validated by the group.

A.4 Sensitivity Analysis

To test whether the main results are driven by a single calibrated constant rather than the underlying structure of the problem, we performed a one-at-a-time sensitivity analysis on three modelling choices: the ARV shortfall penalty P , the ARV–Supply synergy coefficient γ_c (scaled uniformly), and the total budget. For computational tractability, these sensitivity runs used a shorter NSGA-II horizon than the main 300-generation experiment; results should therefore be read as robustness checks, not replacement benchmark results.

Scenario	DALYs	Inequity	ARV %	Non-ARV %	Shortfall	S. Africa	Nigeria
Baseline	4.217M	68,890	81.0	19.0	710	2.8	73.1
$P = 500$	4.582M	66,794	79.9	20.1	712	1.0	73.4
$P = 2000$	3.505M	71,131	83.0	17.0	706	2.2	74.6
γ scale = 0	4.230M	68,607	80.8	19.2	711	0.4	72.7
γ scale = 5	4.236M	66,473	81.1	18.9	710	1.5	74.4
Budget 400M	7.275M	74,812	79.4	20.6	555	0.6	95.4
Budget 800M	11.558M	90,024	78.0	22.0	248	104.2	113.0

Table 5: One-at-a-time sensitivity analysis (NSGA-II knee-point solutions). Shortfall, South Africa allocation, and Nigeria allocation are in USD millions. Because changing P alters the objective penalty itself, DALYs should not be compared directly across different P scenarios; the more meaningful comparisons are intervention shares, shortfall, and allocation pattern. The sensitivity baseline’s South Africa allocation differs from the main 300-generation knee point because the sensitivity runs use a shorter NSGA-II horizon; both results show the same qualitative pattern of near-exclusion under the USD 200M scarcity regime.

The sensitivity analysis supports the main qualitative findings. First, increasing the ARV shortfall penalty P increases the ARV budget share from 79.9% at $P = 500$ to 83.0% at $P = 2000$ (Figure 7), confirming that the penalty is a key driver of ARV concentration, as claimed in Section 8.2. Second, scaling the ARV–Supply synergy coefficient γ_c has little effect: removing synergy entirely (γ scale = 0) leaves the non-ARV share at 19.2%, while a $5\times$ stronger synergy gives 18.9%. This confirms that, under the current units and calibration, the additive synergy term is numerically small relative to the direct DALY response and ARV shortfall penalty. The result should therefore be interpreted as evidence that the implemented synergy proxy is not a material driver of the frontier, rather than as strong evidence about real-world ARV–Supply complementarity. Third, relaxing the total budget has by far the largest structural effect (Figure 6). At USD 800M, the ARV shortfall falls from USD 710M to USD 248M, and South Africa’s knee-point allocation rises from USD 2.8M to USD 104.2M — a roughly 37-fold increase. This directly supports the interpretation in Section 10.3: the near-exclusion of South Africa is largely an artefact of the USD 200M scarcity regime interacting with the saturating response, rather than a fixed property of the model.

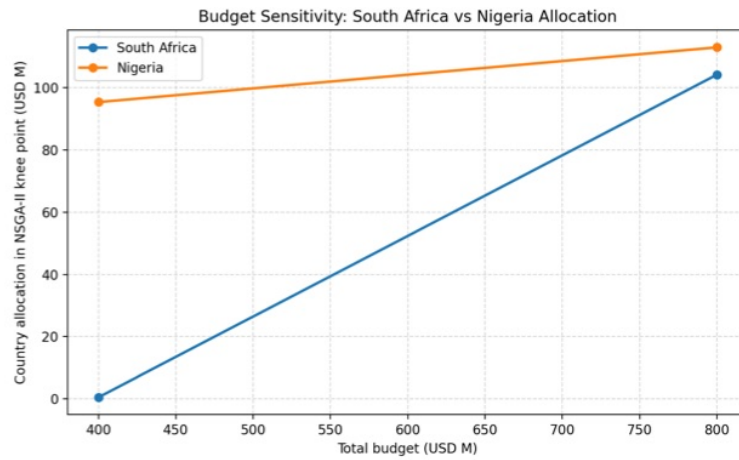


Figure 6: Budget sensitivity for South Africa and Nigeria allocation at the NSGA-II knee point. As the budget increases from USD 400M to USD 800M, South Africa's allocation rises sharply while Nigeria's grows only modestly, indicating that very large-burden countries become fundable once the most severe ARV scarcity is relaxed.

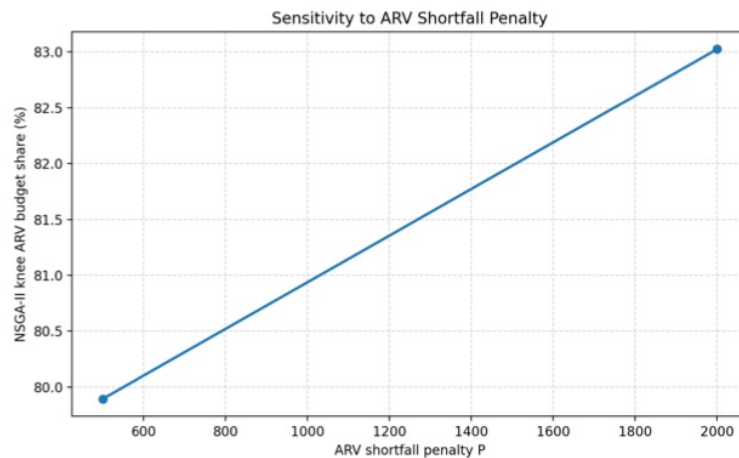


Figure 7: Sensitivity of the NSGA-II knee-point ARV budget share to the shortfall penalty P . A higher penalty pushes more of the budget toward ARV procurement at the expense of Supply, Workforce, and HIVDR.